

Loan Approval Prediction System

1. Project Overview

The objective of this project is to predict whether a loan application will be approved or rejected based on applicant demographic, financial, and credit-related information using Machine Learning techniques.

2. Business Problem

Financial institutions receive numerous loan applications. Manual verification is time-consuming and prone to human bias. This system helps automate the loan approval process by predicting loan eligibility.

3. Dataset Information

The dataset contains applicant information such as:

- Gender
- Marital Status
- Dependents
- Education
- Employment Status
- Applicant Income
- Co-applicant Income
- Loan Amount
- Loan Term
- Credit History
- Property Area

Target Variable:

- Loan_Status (Approved / Rejected)

4. Data Preprocessing

- Missing value handling
- Duplicate removal
- Data cleaning
- Data validation

5. Exploratory Data Analysis

Key findings:

- Applicants with good credit history have a higher approval rate.
- Graduate applicants show better approval rates.
- Semiurban properties have more approved loans.
- Higher total income positively impacts loan approval.

6. Feature Engineering

Created new features:

- TotalIncome
- LoanIncomeRatio
- EMI

7. Model Building

Models Tested:

- Logistic Regression
- Decision Tree
- Random Forest

Selected Model:

- Random Forest Classifier

8. Model Performance

Accuracy: 78.86%

Evaluation Metrics:

- Accuracy Score
- Classification Report
- Confusion Matrix

9. Deployment

The model was deployed using Streamlit to provide real-time loan approval predictions.

10. Conclusion

The Loan Approval Prediction System successfully predicts loan approval status and demonstrates the application of machine learning in financial decision-making.

11. Future Enhancements

- Hyperparameter tuning
- XGBoost implementation
- Cloud deployment
- Integration with banking systems
- Real-time database support